

1. Original Goal State

```
w : ℕ      a b : BitVec w      h : a &&& b = 0#w  
⊢ a + b = a ||| b
```

3. Reflection into Term Language

```
w : ℕ      a b : BitVec w  
⊢ (lor  
  (neq ((var 0).and (var 1)) zero)  
  (eq ((var 0).add (var 1)) (or (var 0) (var 1))))  
  .denote w [a, b]
```

5. FSM Denotation

```
⊢ ∀ (w : ℕ) (a b : BitVec w),  
  <fsm>.eval w [a, b] = BitStream.allOnes
```

2. Negation Normal Form

```
w : ℕ      a b : BitVec w  
⊢ a &&& b ≠ 0#w ∨ a + b = a ||| b
```

4. Bitstream Denotation

```
w : ℕ      a b : BitVec w  
⊢ (lor  
  (neq ((var 0).and (var 1)) zero)  
  (eq ((var 0).add (var 1)) (or (var 0) (var 1))))  
  .eval w [a, b] = BitStream.allOnes
```

6. Deciding FSM Denotation by k-induction

```
w : ℕ      a b : BitVec w  
⊢ proven by <decide>
```